Data Visualization

RevoFinance

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Business Context

RevoFinance is a fictional fintech company that empowers individuals in Indonesia to track and manage their personal expenses across both offline and online merchants. With increased financial literacy and mobile app adoption, **RevoFinance** aims to help users better understand their spending behavior, optimize personal budgets, and take control of their financial health.

Business Goals

- Enable users to track monthly and annual spending trends.
- Visualize spending distribution across categories, locations, and merchants.
- Compare actual spend against budgets.
- Provide actionable insights to improve personal financial planning.

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TL Simulation 1

Worksheet Objective

a. The target user of this dashboard

The target users of this dashboard are **individual RevoFinance app users** who actively manage their personal finances and want to monitor their spending behavior, budget usage, and merchant-level expenses. This dashboard is also relevant for **financial analysts and product teams** to understand user spending patterns and identify opportunities for improving budgeting features in the application.

b. Need this dashboard

Users need this dashboard to:

- Monitor **monthly and annual spending trends**
- Compare **actual expenses versus planned budgets**
- Identify **overspending categories and merchants**
- Understand differences between **online and offline spending behavior**
- Make informed decisions to **optimize personal financial planning**

Without this dashboard, users would lack a consolidated view of their expenses and budget utilization.

c. The dashboard tracks several key metrics

*Total Expense

Measures total spending within the selected period to assess overall financial outflow.

*Total Transactions

Indicates transaction frequency and helps identify spending habits driven by volume.

*Average Expense per Transaction

Helps distinguish whether spending is driven by many small transactions or fewer large ones.

*Spending by Category and Merchant

Highlights over-budget and under-budget categories to support corrective actions.

*Budget Utilization (%)

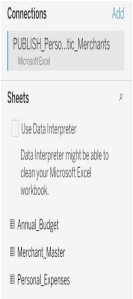
Shows how much of the allocated budget has been used per category.

*Monthly Spending Trend

Identify seasonality, detect spending spikes, and support better budget planning.

Data Preparation

a. Join and blend three data sources



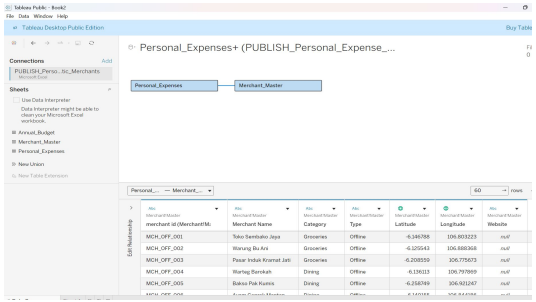
The dashboard integrates three datasets using **Tableau Relationships**:

1. **Personal_Expenses** – transaction-level expense data
2. **Merchant_Master** – merchant metadata including category, type, and location
3. **Annual_Budget** – annual budget allocation per expense category

Relationships are used instead of physical joins to preserve correct aggregation at different levels of analysis.

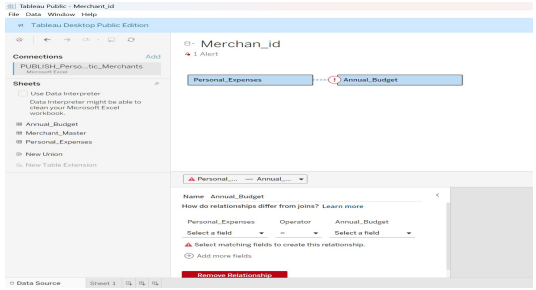
b. Show relationships :

Personal_Expenses ↔ Merchant_Master



- Key: merchant_id
- Cardinality: **Many-to-One**
- Rationale: One merchant can have many expense transactions.

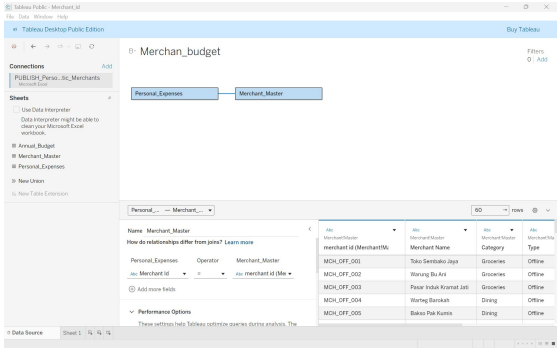
Annual_Budget ↔ Personal_Expenses



This structure enables category-based budget comparison even when the category field is not directly available in Personal_Expenses.

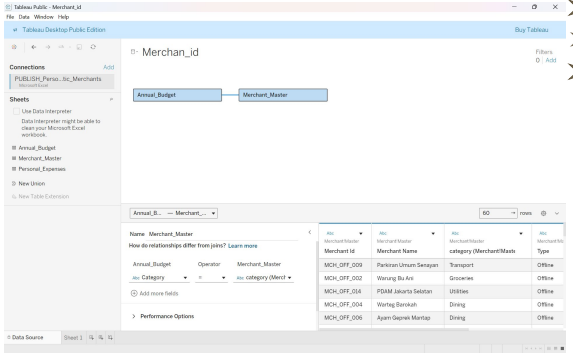
c. Explain the join logic and cardinality :

Cardinality : Expenses ⟶ Merchant



- Key: **merchant_id**
- Cardinality: Many-to-One
- Rationale: One merchant can have many expense transactions.

Cardinality : Merchant ⟶ Budget



- Key: **category**
- Cardinality: Many-to-One
- Rationale: Each category has a single annual budget, while many merchants belong to the same category.

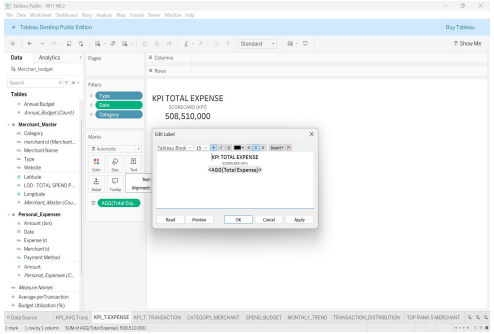
Referential Integrity :

"Set to *Some records match* to account for transactions or merchants that may not have corresponding budget entries."

Basic visualizations

a. Score card : Total Expense, Total Transaction, AVG per Transaction

Total Expense :



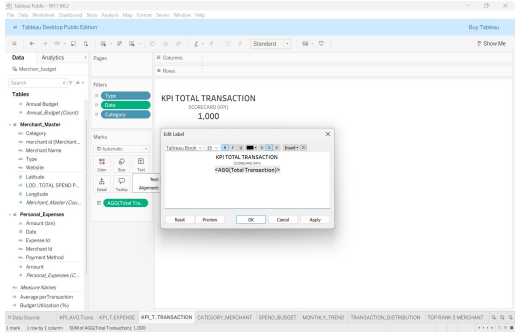
Sheet: KPI_Total_Expense

- Drag **Total Expense** → **Text**
- Format → Currency (IDR)
- Result ***508,510,000***

▼ **create calculated fields**

Total Expense :
SUM([Amount])

Total Transaction :



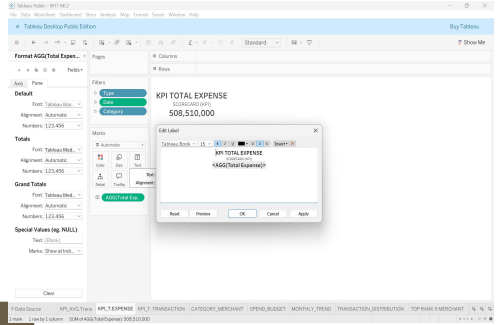
Sheet: KPI_Total_Transaction

- Drag **Total Transaction** → Text
- Result ***1,000*** transaction

▼ **create calculated fields**

Total Transaction :
COUNT([Transaction ID])

Average per Transaction :



Sheet: KPI_Avg_Transaction

- Drag **Average per Transaction** → Text
- Result ***508,510*** (Total Expense / Total Transaction)

▼ **create calculated fields**

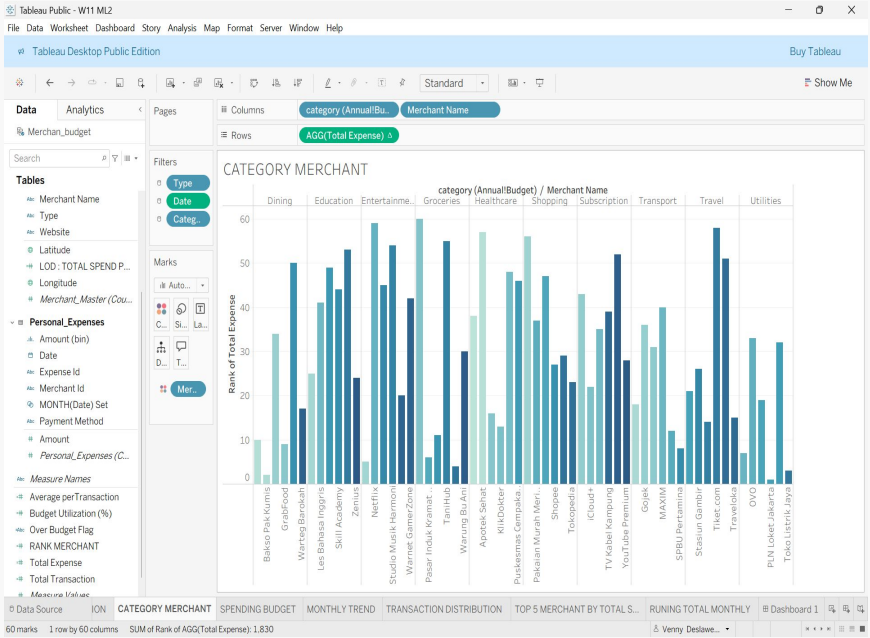
AVG. per Transaction :
AVG([Amount])

Insight:

- **Total Transaction:** ±1.000 transaction
- **Total Expense:** ±508,5 million
- **Average Transaction Value:** ±508 thousand
- Spending is driven by frequent medium-value transactions rather than large one-time expenses.

Basic Visualizations

b. Category hierarchy : Category ⟶ Merchant



Insight :

A hierarchical bar chart allows users to drill down:

- From **Category**
- To **Merchant level**

This supports detailed analysis of spending concentration within categories.

▼Insight :

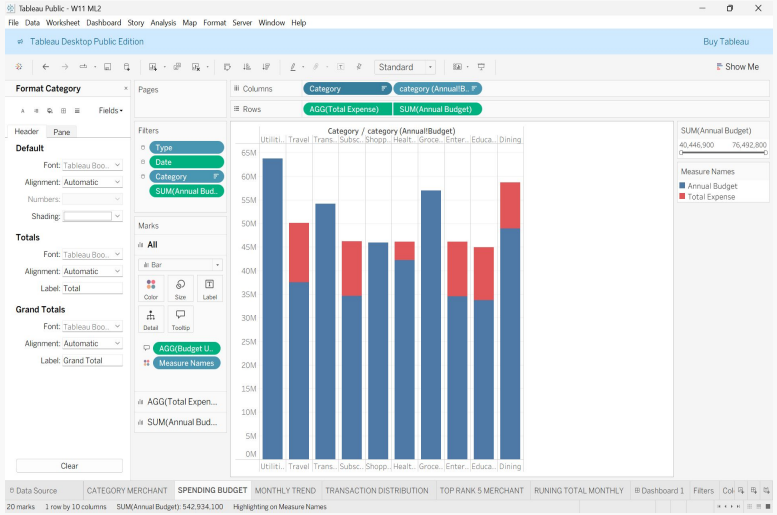
- ❖ In one category, spending is concentrated among a few merchants.
- ❖ Examples: digital services, transportation, and certain groceries dominate.

▼Meaning:

- ❖ Risk of merchant dependency.
- ❖ Optimization opportunities (price comparison/alternatives).

Basic Visualizations

c. Bar chart : Spend per Category vs Budget Utilization (dual axis or bar + label)



Sheet: Spend_vs_Budget

1. Drag **Category** → Columns
2. Drag **SUM(Amount)** → Rows
3. Drag **SUM(Annual Budget)** → Rows
4. Klik kanan axis → **Dual Axis**
5. **Synchronize Axis**
6. Marks:
   - Amount → Bar
   - Budget → Line / Label

▼Insight :

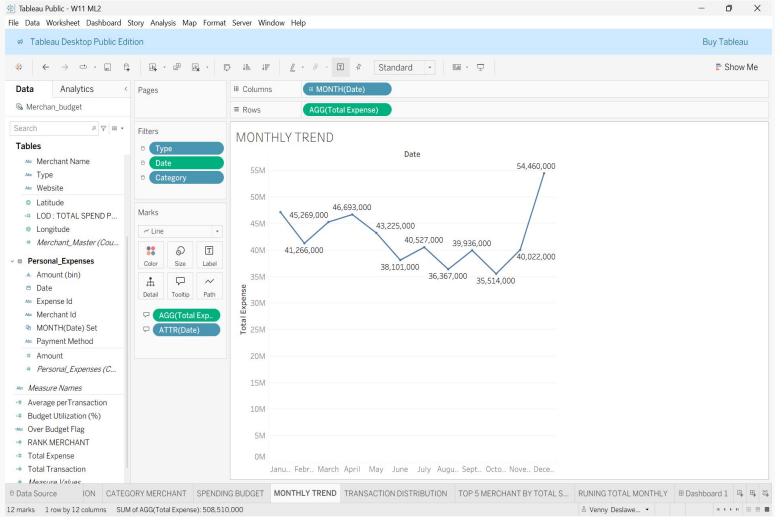
- ❖ In aggregate, total spending is still below the total budget.
- ❖ However, it is not evenly distributed across categories.

▼Meaning:

- ❖ Budget allocation is quite good globally.
- ❖ Control is needed per category, not just the annual total.

Basic Visualizations

d. Line chart : Monthly Total Spending Trend



Sheet: Monthly_Trend

1. Drag **Date** → Columns
2. Klik Date → **Month (Continuous)**
3. Drag **SUM(Amount)** → Rows
4. Marks → **Line**

▼Insight :

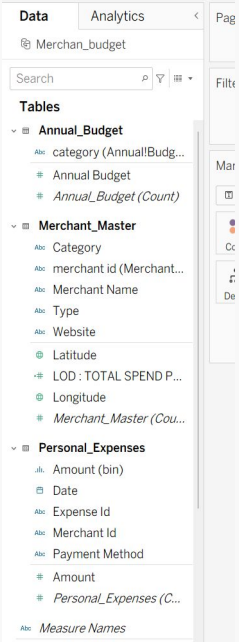
- Fluctuating spending.
- Mid-year decline.
- Significant spike in December.

▼Meaning:

- Indications of seasonality (end of year: holidays, additional needs).
- Risk of budget overruns in Q4.

Basic Visualizations

e. Calculated Field : Budget Utilization (%)



Make **Budget Utilization (%)**

1. Right-click in the **Data Pane**
2. Select **Create → Calculated Field**
3. Enter **THIS formula**

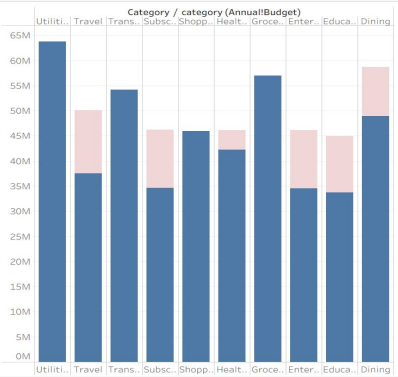
Budget Utilization

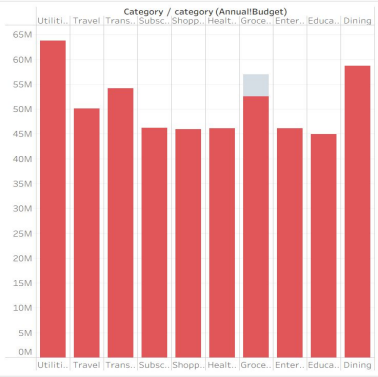
```
SUM([Total Expense]) /
SUM([Annual Budget])
```

4. Click → **Oke**

Make **Percentage (%)**

1. Right-click in the field **Budget Utilization**
2. Select **Deafault Properties → Number Format**
3. Select **Percentage**
4. Decimal : 1-2 number





▼Insight :

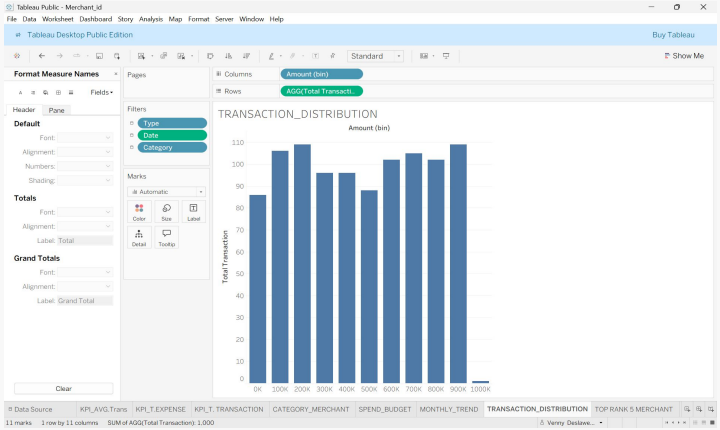
- Budget utilization ranges from ±76% – 111%
- Some categories exceed 100%

▼Meaning:

- The budget is not adaptable to real spending behavior
- The initial allocation is unrealistic for certain categories

Basic Visualizations

f. Bin Chart : Distribution of transaction amounts (e.g <100k, 100-250k, etc.)



Make BIN

Klik kanan Amount → **Create** → **Bins**

- **Size : 100000**

Sheet: Transaction_Distribution

1. Drag **Amount BIN** → Columns
2. Drag **Count(Amount)** → Rows
3. Marks → Bar

▼Insight :

- ❖ The majority of transactions are in the 100,000-900,000 range.
- ❖ There are almost no extreme transactions above 1 million.

▼Meaning:

- ❖ Stable spending patterns
- ❖ Financial risk comes from volume, not extreme amounts.

Basic visualizations

g. The Visualization presented by the fundamentals of visualization? A better form of visualization than the one requested?

All visualizations follow data visualization best practices:

- Appropriate chart types for each metric
  Consistent color usage
- Clear labeling and readable axes
- Reduced visual clutter

Where appropriate, bar charts and line charts were chosen over more complex visuals to improve clarity and interpretability.

TL Simulation 2

Map Visualization

Sheet: **Offline_Map**

1. Filter **Location = Offline**
2. Drag **Longitude** → Columns
3. Drag **Latitude** → Rows
4. Marks → **Circle**
5. Size → **SUM(Amount)**
6. Color → **Category**
7. Filter city → Jakarta

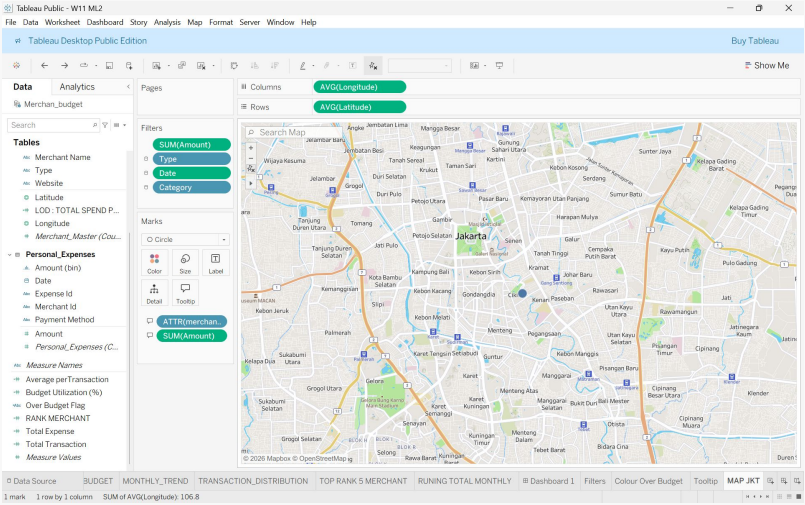
▼ Insight :

- Offline transactions are concentrated in the Jakarta area.
- Not spread nationwide.

▼ Meaning:

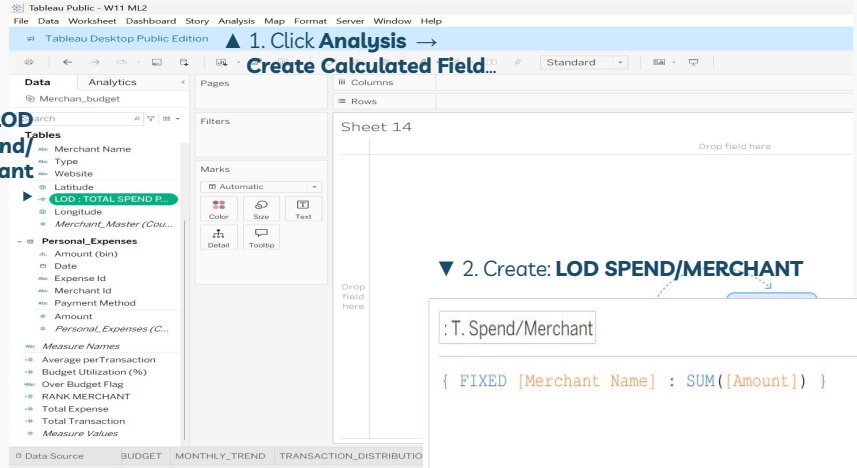
- Offline spending is local.
- Potential for optimizing local merchants/loyalty programs.

▼ Map Offline Jakarta



Advanced Calculation

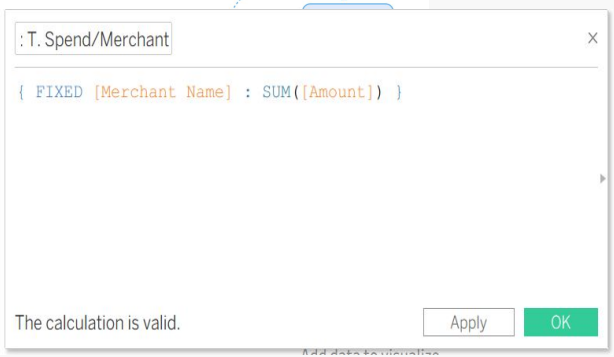
▼ LOD: Total Spend/Merchant



LOD Spend/ Merchant

3. Create: { FIXED [Merchant Name] : SUM([Amount]) }▸

▼ 2. Create: **LOD SPEND/MERCHANT**



◂ 4. Click **Ok**

▼Insight :

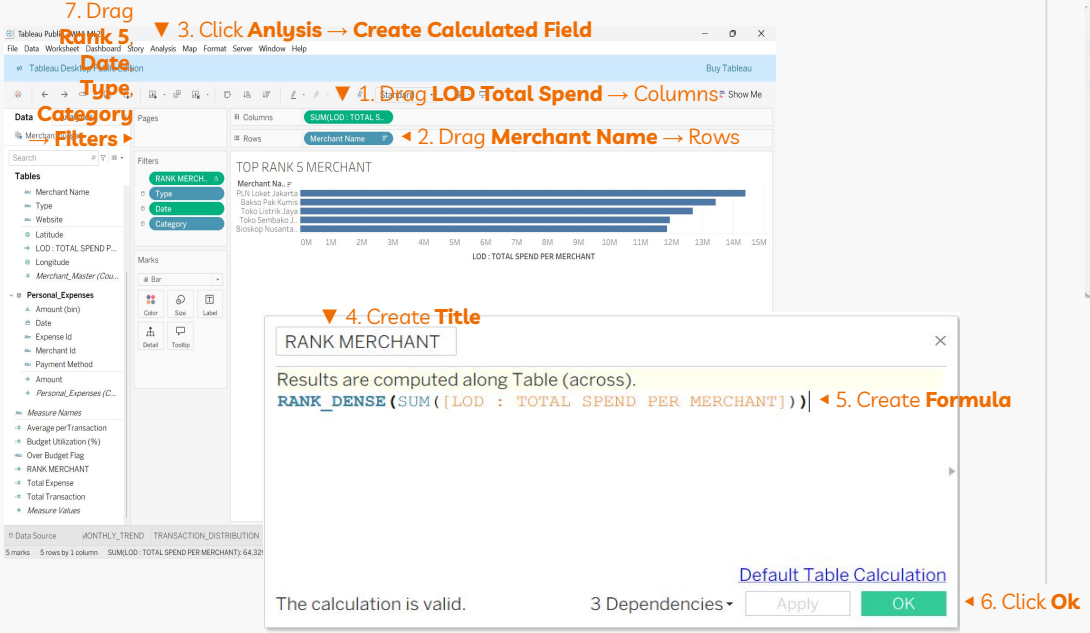
- ❖ **LOD Total Spending per Merchant** calculates the total expense for each merchant across all available time periods using a FIXED level of detail.

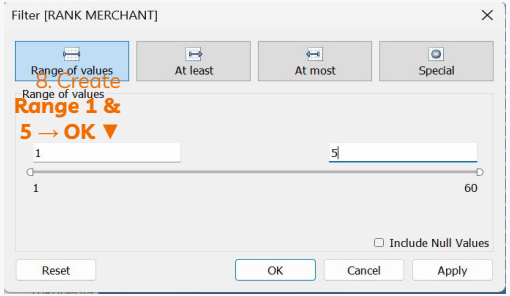
▼Meaning:

- ❖ This allows a consistent comparison of merchant contribution to total spending, independent of date filters. It helps identify high-impact merchants, spending concentration, and long-term cost optimization opportunities..

Advanced Calculation

▼ Ranking Top 5 Merchant by Total Amount





▼Insight :

- ❖ The top 5 merchants contributed a significant portion of total spending.
- ❖ There are utility and routine needs merchants.

▼Meaning:

- ❖ High spending is not an anomaly, but structural.
- ❖ Suitable for negotiation, bundling, or limit control.

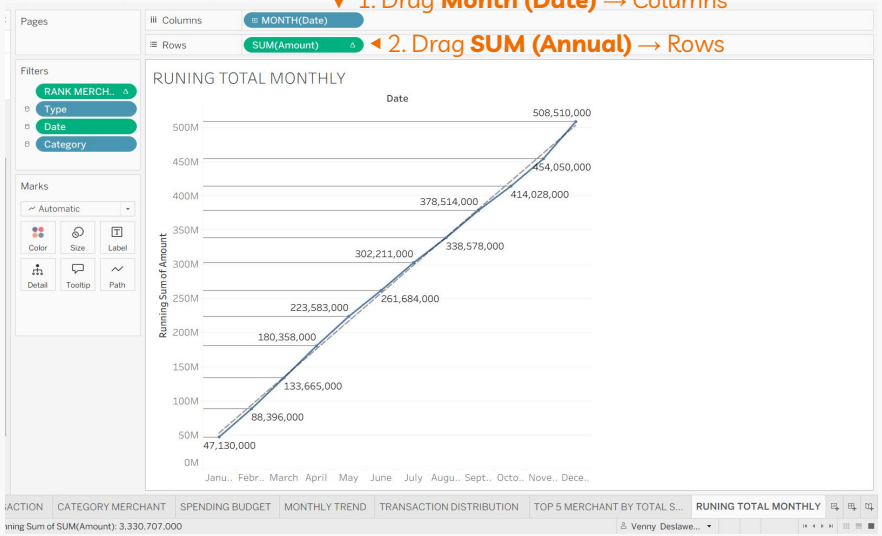
Advanced Calculation

▼ Running Total Monthly

3. Drag **Rank 5, Date, Type, Category → Filters** ▸

▼ 1. Drag **Month (Date)** → Columns

◂ 2. Drag **SUM (Annual)** → Rows



▼Insight :

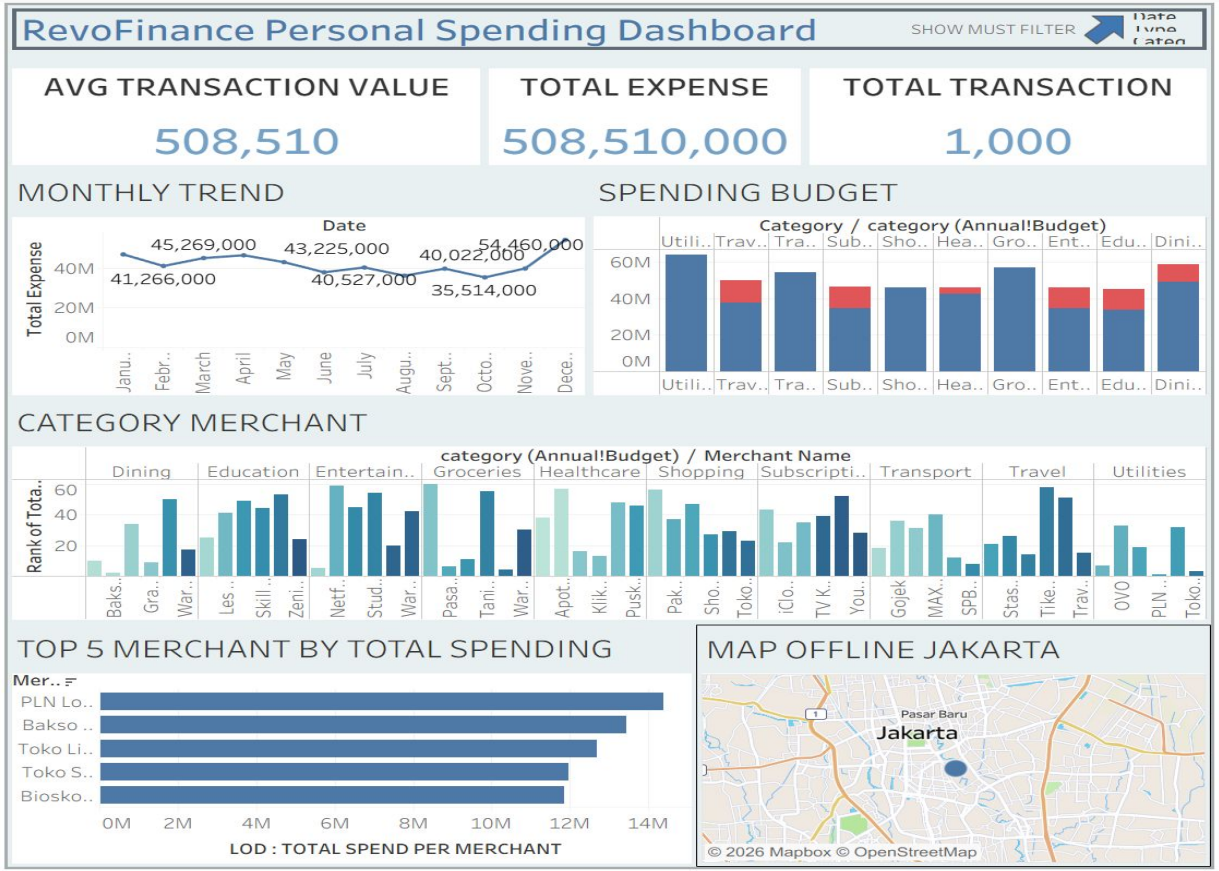
- ❖ Stable cumulative increase throughout the year
- ❖ Sharp increase at the end of the year

▼Meaning:

- ❖ Monthly control is important
- ❖ Without early warning, the end of the year is prone to overspending

Dashboard

Link to Tableau Public



Key Insights & Recommendations OBIPR

1. Spending Pattern Is Volume-Driven

- **O (Observation)**
  Most transactions fall within the medium-value range (100K–900K), with very few high-value transactions.
- **B (Business Impact)**
  Total expenses increase primarily due to transaction frequency rather than transaction size.
- **I (Insight)**
  Frequent small-to-medium transactions create cumulative overspending risk.
- **P (Proposal)**
  Introduce monthly spending caps and transaction frequency monitoring.
- **R (Result)**
  Improved expense control without restricting essential high-value purchases.

2. Category-Level Budget Imbalance

- **O (Observation)**
  While overall spending remains within the total annual budget, several categories exceed their allocated budgets.
- **B (Business Impact)**
  Category-level overspending reduces flexibility and increases financial risk.
- **I (Insight)**
  Current budget allocation does not fully reflect actual spending behavior by category.
- **P (Proposal)**
  Reallocate budget based on historical spending patterns and adjust category limits dynamically.
- **R (Result)**
  More realistic budgets and reduced over-budget incidents.

Key Insights & Recommendations OBIPR

3. Merchant Spending Concentration

- **O (Observation)**
  Top 5 merchants contribute a significant portion of total expenses across all time.
- **B (Business Impact)**
  High dependency on a small number of merchants increases cost exposure.
- **I (Insight)**
  Optimizing spending with a few high-impact merchants can generate substantial savings.
- **P (Proposal)**
  Negotiate pricing, switch providers, or consolidate services for top-spending merchants.
- **R (Result)**
  Lower long-term costs and improved spending efficiency.

4. Seasonal Spending Risk

- **O (Observation)**
  Monthly spending shows a noticeable spike toward the end of the year.
- **B (Business Impact)**
  Without early controls, seasonal spikes can lead to unexpected budget overruns.
- **I (Insight)**
  Spending behavior is strongly influenced by seasonal factors.
- **P (Proposal)**
  Apply seasonal budgeting and early warning alerts when budget utilization exceeds 80%.
- **R (Result)**
  Better anticipation and control of year-end expenses.

Key Insights & Recommendations OBIPR

5. Offline Spending Is Location-Concentrated

- **O (Observation)**
  Offline transactions are highly concentrated in Jakarta.
- **B (Business Impact)**
  Limited geographic spread suggests localized spending behavior.
- **I (Insight)**
  Location-based optimization strategies can be effective.
- **P (Proposal)**
  Introduce location-based promotions or merchant partnerships in high-spending areas.
- **R (Result)**
  Increased value from offline spending without increasing total expenses.

Thanks!